Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/06/2025
PI Name:	Macaulay, Timothy Ph.D.		
Project Title:	Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Macaulay)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3711	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	01/30/2019	**End Date:**	09/30/2033
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Project start date entered as 01/30/2019 to match original investigation, as requested by NASA HHC (Ed., 1/3/25).		
Key Personnel Changes/Previous PI:	September 2024 Update: The new Principal Investigator for this project will be Timothy Macaulay, Ph.D. The previous Principal Investigator, Dr. Gilles Clement, is retiring. Dr. Clement will continue as a Co-Investigator (Ed., 1/2/25).		
COI Name (Institution):	Dervay, Joseph M.D. (NASA Johnson Space Center) Makishima, Tomoko M.D., Ph.D. (University of Texas Medical Branch at Galveston) Wood, Scott Ph.D. (NASA Johnson Space Center) Clement, Gilles Ph.D. (KBR/NASA Johnson Space Center - Retired)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Adaptation to the absence of Earth's gravitational environment during spaceflight causes neurological disturbances that are either directly or indirectly mediated by the vestibular system. These disturbances include space motion sickness, spatial disorientation, and cognitive impairment, as well as changes in head-eye coordination, vestibulo-ocular reflex, and interactions with support surfaces. After return to Earth, astronauts experience other vestibular-driven behavioral changes, including re-entry motion sickness, motion illusions, gaze-induced nystagmus, and balance and locomotion deficits. Otolith-mediated reflex gain changes are striking shortly after g-transitions. However, animal studies have shown that structural modifications of the vestibular sensory apparatus may occur throughout an extended spaceflight exposure. To date, no long-duration flight studies have directly investigated potential changes in the vestibular organs of astronauts. As a part of the Complement of Integrated Protocols for Human Research (CIPHER), this study will address this gap by performing a systematic neuro-vestibular examination of crewmembers in orbit at regular intervals, as well as immediately after landing. Crewmembers of various mission lengths will be recruited to investigate temporal changes, and to identify trends in adaptation to vestibular health and performance. In orbit, subjects will perform active head movements and be exposed to various passive maneuvers executed by the operator. Observations and recordings of eye, head, and body movements, as well as subjective perception of motion and verbal reports, will be used for evaluating the presence of abnormal eye movements, dysmetria, motion sickness symptoms, and illusions of motion during head or body movements. Tests will be performed both early in the mission and once every one or two months thereafter. The post-flight examination will be performed on R+0 and several times thereafter. The tests proposed in this study are well established and validated protocols that can detect acute or chronic vestibular syndromes. Repeated measurements during adaptation to g-transitions will provide insight into the vestibular adaptation processes. If the observed physiological changes in crewmembers are more deleterious after longer duration missions (1+ years) on the International Space Station (ISS) than those documented after standard-duration (6 month) ISS expeditions, then relevant countermeasures will be required to enable longer duration missions. Depending on the etiology of the vestibular disorders revealed by our tests, monitoring for long-term health outcomes and vestibular rehabilitation countermeasures can be tailored to the deficits observed. During one preflight session, a simple and rapid measure of otolith asymmetry – a basic low-level neurovestibular function that is altered in spaceflight – will also be administered. This provides a measure of binocular alignment (vertical and torsional), which reflects underlying otolith asymmetry. This testing provides an opportunity to assess a low-level neural system that is not under volitional control, is responsive to g-level alterations, and might be an indicator of broader and more functionally relevant effects including inflight/postflight measures of (mal)adaptation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The tests proposed in this study are well established and validated protocols that can detect acute or chronic vestibular syndromes. Repeated measurements during adaptation to g-transitions will provide insight into the vestibular adaptation processes. If the observed physiological changes in crewmembers are more deleterious after longer duration missions (1+ years) on the International Space Station (ISS) than those documented after standard-duration (6 month) ISS expeditions, then relevant countermeasures will be required to enable longer duration missions. Depending on the etiology of the vestibular disorders revealed by our tests, monitoring for long-term health outcomes and vestibular rehabilitation countermeasures can be tailored to the deficits observed.
Task Progress:	Ed. Note - September 2024: Dr. Clement is retiring from NASA; the new PI is Timothy Macaulay, Ph.D. For additional information on this investigation, see: 1) "Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Reschke)": https:// . From 2019-2022, the Principal Investigator was Millard Reschke, Ph.D. 2) "Neuro-Vestibular Examination During and After Spaceflight (Vestibular Health) (PI: Clement)": https:// . From 2022-2024, the Principal Investigator was Gilles Clement, Ph.D. ---------- This past reporting period, the study team continued collecting data on the International Space Station (ISS) crewmembers, as part of CIPHER. These included 2 preflight, 1 inflight, and 4 postflight data collections. These were accomplished despite the challenges of spaceflight operations. To date, three crewmembers have completed all preflight, inflight, and postflight testing for this study. An inflight functional checkout was completed to identify causes and mitigations for unexpected noise in the video eye-tracking data collected during inflight data collection. The study team worked with ISS hardware engineers, the device manufacturer, and one crewmember on the ISS for this checkout. While wearing the goggles, the crewmember, a physician, indicated that the noise was likely driven by their heart rate, suggesting that the external arteries don't have as much autoregulatory control in a microgravity environment. This was confirmed by capturing a stable video with the goggles doffed. Loosening the goggle straps relieved some of the pressure and noise. This checkout has resulted in novel findings related to headward fluid shifts in a microgravity environment and procedural mitigations for future inflight data collections. The inflight and post-flight Motion Sickness Questionnaires (MSQs) have been updated again, coordinating across three different studies to optimize crew time efficiency and sustain support for administration of this critical sensorimotor outcome measure. The Flight Day 1 (FD1) MSQ will only be completed if the subject is unable to complete the Vestibular Exam. The FD3 MSQ will be administered by and shared from NASA Human Research Program's (HRP's) Spaceflight Standard Measures. We collaborated with the Research Operations and Integration (ROI) team and EveryWear application developers to format the FD3 MSQ tool with the desired survey software logic and branching. The R+0 MSQ has been removed since data will be captured in the R+4 MSQ, which has also been updated to match the R+4 MSQ from HRP's Spaceflight Standard Measures. This R+4 MSQ is designed to be completed using a secure survey software link or physical paper copy, which we or our ROI partners will provide to the crew. Standardizing these

processes across multiple studies will enhance future data mining efforts to analyze measures for evaluating countermeasure efficacy.

Progress has been made on the development of automated data processing pipelines for eye tracking data. Several analysis routines were consolidated into a single Matlab application with a unified Graphical User Interface (GUI). This development allows the comparison of different eye tracking processing algorithms in a modular approach that can be shared across the different study-specific aims. The modular approach increases the efficiency with which new analysis tools can be implemented by building on the same calibration and GUI framework.

We previously reported normative data from thirty-two healthy ground (non-astronaut) volunteers who performed the same tasks as our crew subjects (18 females, 14 males; 38.6 ± 9.2 years). We also previously showed that astronauts early postflight exhibit dynamic balance control functions comparable to those of individuals with bilateral vestibulopathy (BVP) (Clement et al., 2023). Recently, thirty patients with BVP at the University of Caen (17 females, 13 males; 60.6 ± 13.0 years) performed a subset of our crew subject tests using the same protocols. These tests involved the perception of egocentric distance during blinded walking, perception of egocentric rotation during blinded turning, and path integration during blinded walking in a Triangle Completion Task. While performing all these tasks, the subjects wore external noise-canceling earphones that masked any auditory clues of spatial orientation. Patients with BVP showed significantly greater errors in judging distances during blind walking ($p < 0.05$) than the healthy ground volunteers; however, their errors in judging rotations during blind turning were not significantly different from the healthy volunteers, except for small turn angles (i.e., 90 degrees). During the path integration task, both distance error and angle of deviation differed significantly between groups ($p < 0.05$). These data suggest that vestibular deficits impact all aspects of body movement perception tested; however, other systems (i.e., proprioception and cognitive strategies) can help individuals compensate in some tasks, like separately perceiving distances and rotation. These findings are promising for exploration spaceflight because transitions between gravitational environments mimic the loss of vestibular otolith function. Comparisons will be made with data collected from crewmembers performing the same tasks after their returns from spaceflight.

Bibliography Type:	Description: (Last Updated: 12/23/2025)
Abstracts for Journals and Proceedings	Macaulay TR, Kuldavletova O, Denise P, Wood SJ, Reschke MF, De Dios YE, Dervay JP, Makishima T, Schubert MC, Shelhamer M, Kheradmand A, Clement GR. "Neuro-vestibular examination during and following spaceflight (Vestibular Health)." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025